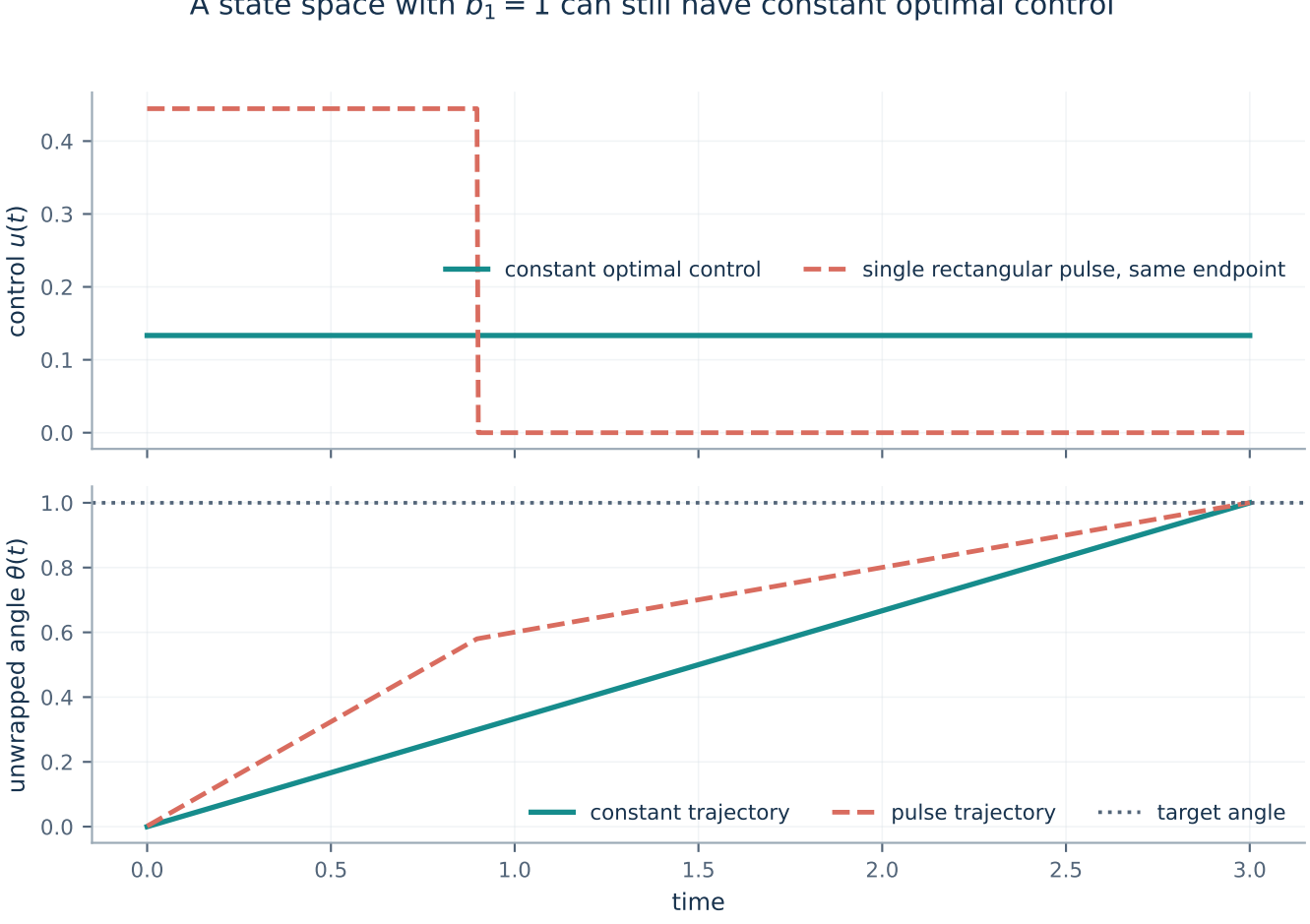
A state space with $b_1 = 1$ can still have constant optimal control
control $u(t)$
0.0
0.1
0.2
0.3
0.4
constant optimal control
single rectangular pulse, same endpoint
unwrapped angle $\theta(t)$
0.0
0.2
0.4
0.6
0.8
1.0
constant trajectory
pulse trajectory
target angle
0.0
0.5
1.0
1.5
2.0
2.5
3.0
time